Yitong Bai

M.S. Student in Computer Engineering, Columbia University

New York, NY   yb2636@columbia.edu   (646) 221-1046   brianbai093.github.io   BrianBai093

brianbai

Research Interests

- LLM agents, multi-agent systems, ML systems, evaluation and verification, reproducibility, trustworthy AI, and LLM post-training.

Education

M.S. **Columbia University**, Computer Engineering — New York, NY

Sept 2025 – Dec 2026
- Expected Dec 2026
- GPA: 3.83 / 4.00

B.S. (Hons) **Lancaster University**, Computer Science — Lancaster, UK

Sept 2023 – June 2025
- First Class Honours; GPA: 3.73 / 4.00
- BJTU-Lancaster Dual-Degree Program (2+2)

B.S. **Beijing Jiaotong University**, Computer Science and Technology — Beijing, China

Sept 2021 – June 2023
- GPA: 3.37 / 4.00
- BJTU-Lancaster Dual-Degree Program (2+2)

Research Experience

National Zhijiang Lab, Research Assistant — Hangzhou, China

Advisor: Jiaxi Yang, Research Scientist, National Zhijiang Lab — June 2026 – present

- Project: Feedback Learning for EHR SQL Repair: SFT vs. GRPO; manuscript in preparation.
- Conducted a pre-registered study comparing SFT and GRPO on how diagnostic feedback shapes learning for EHR SQL repair, fine-tuning Qwen3.5-9B on MIMIC-IV EHR questions under four nested feedback levels, from a bare failure signal to repair hints.
- Trained 12 QLoRA adapters across 4 feedback conditions x 3 seeds, and evaluated semantic repair success across 13 model groups and 10,400 records with task-clustered confidence intervals, permutation tests, and Holm correction.
- Found no stable incremental benefit from richer feedback under SFT: matched gains of +1.0 to +2.0 percentage points did not meet the pre-registered significance bar.
- Built a target-free GRPO pipeline with a custom binary execution reward and observed a non-monotonic feedback effect, with the intermediate L2 level reaching +10.0 points (95% CI [+4.0, +16.5], exploratory Holm-adjusted $p = 0.0228$), pending multi-seed confirmation.

Columbia University, CRIS Lab, Research Assistant — New York, NY
Advisor: Prof. Venkat Venkatasubramanian — Sept 2025 – present

- Project: DeepAudit: Agentic Evidence-Based Auditing of ML Reproducibility; manuscript in preparation.
- Designed and implemented DeepAudit, an agentic claim-level reproducibility auditing system that maps ML paper claims to executable repository tasks, metric contracts, runtime artifacts, and structured verdicts.
- Built a 23-step Python pipeline covering PDF-to-Markdown ingestion, table/figure-aware claim extraction, repository analysis, RAG-assisted code indexing, environment repair, autonomous execution, evidence alignment, scoring, and report generation.
- Formulated ML reproducibility auditing as claim-level evidence alignment rather than binary repository execution, distinguishing metric disagreement from missing data, incompatible environments, ambiguous entry points, unlogged metrics, and compute limits.
- Evaluated DeepAudit on 10 public ML paper-code pairs with 64 pre-specified audit targets; extracted 1,101 claims, identified 988 code-verifiable claims (89.7%), generated 49 metric contracts, and executed 52/64 experiment specifications (81.2%).

Columbia University / IBM Research Collaboration, Research Assistant — New York, NY
Mentored by Dhaval Patel, Senior Technical Staff Member, IBM Research — Feb 2026 – present

- Project: Multi-Turn AssetOps: Supervisor-Specialist Agents for Industrial O&M.
- Proposed and designed a LangGraph-based Supervisor-Specialist multi-agent architecture for multi-turn industrial asset operations and maintenance.
- Designed structured cross-turn artifact reuse for sensor evidence, anomaly analyses, failure hypotheses, and maintenance recommendations.
- Optimized routing, artifact reuse, and MCP execution with constraints, schema validation, duplicate-call suppression, and failure guards.
- Co-developed and evaluated 16 multi-turn industrial dialogs, reporting gains in planning effectiveness, task completion, tool-time share, and follow-up latency after artifact reuse.

SJTU Summer Campus, Project Leader — Shanghai, China
Facial Emotion Recognition with VGG16 — July 2023 – Sept 2023

- Built a facial emotion recognition pipeline on the FER-2013 dataset using VGG16-based feature extraction and an MLP classifier.
- Applied data augmentation and feature fusion to improve robustness across emotion categories.
- Led a four-person team, coordinated task division and progress reviews, and completed both implementation and project reports.

Publications and Manuscripts

Towards Multi-Turn Dialog Systems for Industrial Asset Operations and Maintenance — 2026
Chengrui Li, Rujing Li, *Yitong Bai*, Rui Li
arxiv.org/abs/2605.24953 (2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); arXiv:2605.24953)

Application and Analysis of the VGG16 Model in Facial Emotion Recognition — 2023
Yitong Bai
10.5220/0012799800003885 (Proceedings of DAML 2023, pp. 422-426)

Position: Agentic Science Needs Explicit Claim-Result-Computation Relationships — 2026
Yitong Bai
(Manuscript under review)

Beyond Static Leaderboards: Predictive Validity for the Evaluation of LLM Agents 2026
Dhaval Patel, *Yitong Bai*, et al.
(Manuscript under review)

The Label Is Not the Answer: Output-Schema Anchoring in Medical Multimodal MCQ Evaluation 2026
Chengyi Zhang, *Yitong Bai*, Ziyang Wang
(Accepted at MICCAI 2026 Challenge MedReason)

Industry Experience

Neusoft Group, Software Engineering Intern — Guangzhou, China
July 2024 – Aug 2024

- Contributed to development and testing for a university portal system, including frontend refinement, database interface support, and cross-database data matching.
- Worked with heterogeneous data sources to support synchronization and integration across subsystems.

Selected Projects

Agentic Java Formal Verification — Dec 2025
Columbia University, Formal Verification Project

- Built an agentic Java verification assistant that translates natural-language proof goals into Java annotations and iteratively repairs verifier errors.
- Designed a verification loop over OpenJML feedback, with prompt construction, error parsing, retry control, and proof-state tracking.

ConvNeXt and ViT for Hierarchical Open-Set Image Classification — Dec 2025
Columbia University, Neural Networks and Deep Learning Project

- Built a two-stage hierarchical open-set recognition pipeline in PyTorch for super-class and sub-class prediction.
- Implemented interchangeable ImageNet-pretrained ConvNeXt and ViT backbones for controlled comparison.
- Evaluated on a 6,288-image dataset with 3 super-classes and 87 sub-classes.
- Improved unseen sub-class recognition from 62.24% to 82.22% using ConvNeXt on a held-out novel split.

TrueShot — 2026
Columbia University, Computer Networks Project

- Built a Dockerized Flask LAN game service with screenshot submission, peer validation, blockchain-style logging, and tamper-evident replay.
- Implemented and tested blockchain, networking, application, and web layers.

Computer Vision and ML Lead, Pet Companion Robot — Dec 2025
Columbia University, AIoT Project

- Led computer vision and machine learning components for an AIoT pet robot integrating on-device perception, motion control, cloud analysis, and dashboarding.
- Built real-time dog detection and motion control on Raspberry Pi 5 using YOLO, FastAPI, sensor fusion, and target tracking.
- Developed cloud pipelines for automated image capture, behavior summarization, and a VGG16-based pet emotion classifier.

Honors and Awards

- Honorable Mention, Mathematical Contest in Modeling, COMAP (Mar 2024)
- Beijing Jiaotong University Scholarship for Academic Progress, Top 5% (Jun 2022)

Skills

Programming: Python, C, Java, R, SQL, HTML

ML / LLM Tools: PyTorch, scikit-learn, NumPy, pandas, LangGraph, LangChain, RAG, MCP / FastMCP, LiteLLM, Pydantic, OpenAI API, IBM WatsonX

Systems / Engineering: Linux, Docker, Git, Conda / mamba / venv, pytest, Flask, FastAPI